



COS 332 S1 2020

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Review Test Submission: Semester test 2

User Mr S (Shivek) S Goodoon
Course COS 332 S1 2020
Test Semester test 2
Started 6/23/20 4:13 PM
Submitted 6/23/20 5:49 PM
Due Date 6/23/20 7:30 PM
Status Completed
Attempt Score 35 out of 50 points
Time Elapsed 1 hour, 36 minutes out of 2 hours
Instructions

- COS332 Semester Test 2 (23 June 2020)
- This is a two hour test to be written at any time in the window from 15:30 to 19:30.
- Note that the time allocated is longer than what would typically have been allocated, given the challenges of writing the test in a context where you may encounter connectivity problems and other technical challenges.
- This is a closed-book test. You are not allowed to obtain help from any person or other source while writing the test. You are, however, allowed to email the lecturer during the test at molivier@cs.up.ac.za if anything is unclear.
- You have two hours to complete the test after you started writing. However, all writing will be suspended at 19:30, even if you started writing less than two hours before that time.

Results Submitted Answers, Correct Answers
Displayed

Question 1

0 out of 0 points

Do you promise not to consult any prohibited resources while writing this test?

Selected Answer: ☒ True

Correct Answer: ☒ True

Question 2

1 out of 1 points

For this and other multiple choice questions in the paper, select the alternative that fits the question best in each case.

The routing prefix in an IPv6 address (normally) consists of the ... of the address.

Selected Answer: ☒ First 64 bits

Correct Answer: ☒ First 64 bits

Question 3

1 out of 1 points

The whois command should (eventually) query the ... records.

Selected Answer: ☒ Registrar's

Correct Answer: ☒ Registrar's

Question 4

1 out of 1 points

When a message moves through the protocol stack towards the physical layer, headers are

Selected Answer: ☒ Added

Correct Answer: ☒ Added

Question 5

1 out of 1 points

Multicast (class D) IP addresses may be used to establish

Selected Answer: ☒ None of the above

Correct Answer: ☒ None of the above

Question 6

1 out of 1 points

Which of the following is (are) not a function of ISO/OSI's layer 2?

Selected Answer: ☒ Dialogue control

Correct Answer: ☒ Dialogue control

Question 7

1 out of 1 points

Which of the following is not a valid second-level domain in the za ccTLD?

Selected Answer: ☒ com

Correct Answer: ☒ com

Question 8

1 out of 1 points

Layer 6 of ISO/OSI is the ... layer.

Selected Answer: ☒ Presentation

Correct Answer: ☒ Presentation

Question 9

Question 9 Which of the following URLs is/are valid? Selected Answer: ☒ More than one of the above Correct Answer: ☒ More than one of the above	1 out of 1 points
Question 10 Which of the following infrastructure is inherently able to communicate in full-duplex? Selected Answer: ☒ RS-232 Correct Answer: ☒ RS-232	1 out of 1 points
Question 11 ARP translates a(n) ... address to a(n) ... address. Selected Answer: ☒ IP, MAC Correct Answer: ☒ IP, MAC	1 out of 1 points
Question 12 Suppose a node using SDLC has to transmit the data 01111110. The data will be observed on the line as Selected Answer: ☒ 011111010 Correct Answer: ☒ 011111010	1 out of 1 points
Question 13 To which ISO OSI layer does TCP/IP's Internet layer map? Selected Answer: ☒ 4 Correct Answer: ☒ 3	0 out of 1 points
Question 14 ADSL Selected Answer: ☒ More than one of the above Correct Answer: ☒ More than one of the above	1 out of 1 points
Question 15 MACA is an example of Selected Answer: ☒ Packet passing Correct Answer: ☒ CSMA/CA	0 out of 1 points
Question 16 The end of the payload of an Ethernet (802.3) frame is indicated by a Selected Answer: ☒ Size field in the header Correct Answer: ☒ Size field in the header	1 out of 1 points
Question 17 Which of the following is not a valid DNS record type? Selected Answer: ☒ AA Correct Answer: ☒ AA	1 out of 1 points
Question 18 A zone file with serial number 2001102002, according to convention, was probably created on Selected Answer: ☒ 20 October 2001 Correct Answer: ☒ 20 October 2001	1 out of 1 points
Question 19 Which of the following is not a valid nslookup command? Selected Answer: ☒ resolve up.ac.za Correct Answer: ☒ resolve up.ac.za	1 out of 1 points
Question 20 The Ethernet card with the MAC address 00:50:DA:45:B5:E8 was manufactured by 3Com. Which of the following cards was/were therefore also manufactured by 3Com? Selected Answer: ☒ 00:50:DA:06:47:72 Correct Answer: ☒ 00:50:DA:06:47:72	1 out of 1 points
Question 21 WiFi uses a logical ... topology. Selected Answer: ☒ Star Correct Answer: ☒ Bus	0 out of 1 points

Question 22 Ethernet uses ... as its access methodology. Selected Answer: CSMA/CD Correct Answer: CSMA/CD	1 out of 1 points						
Question 23 Latency refers to Selected Answer: The time it takes to process and transmit a frame. Correct Answer: The time it takes to process and transmit a frame.	1 out of 1 points						
Question 24 X.509 is a standard of the Selected Answer: IEEE Correct Answer: ITU	0 out of 1 points						
Question 25 The broadcast address on the network containing the host 148.231.102.11/12 (assuming that the network is not subnetted) is Selected Answer: 148.231.255.255 Correct Answer: 148.239.255.255	0 out of 1 points						
Question 26 Three stations are waiting to transmit data on a medium where 5-persistent CSMA/CD is used. What is the probability of a successful transmission the moment when the medium becomes available? Selected Answer: 3/8 Correct Answer: 3/8	1 out of 1 points						
Question 27 Which ISO OSI layer may provide general network functionality that resembles the notion of a transaction in a database system? Selected Answer: 3 Correct Answer: 5	0 out of 1 points						
Question 28 What is specified by X.400? Selected Answer: E-mail Correct Answer: E-mail	1 out of 1 points						
Question 29 The encryption algorithm intended to replace DES is Selected Answer: AES Correct Answer: AES	1 out of 1 points						
Question 30 Ethernet addresses have to be unique. This is accomplished as follows: Selected Answer: The IEEE assigns a unique manufacturer's code to each manufacturer. The manufacturer then uses this code as the first half of the address and assigns a unique number to each card manufactured by it that is then used as the second half of the address. Thus a unique address is formed. Correct Answer: The IEEE assigns a unique manufacturer's code to each manufacturer. The manufacturer then uses this code as the first half of the address and assigns a unique number to each card manufactured by it that is then used as the second half of the address. Thus a unique address is formed.	1 out of 1 points						
Question 31 CSMA/CA (Collision Avoidance) Selected Answer: Virtually eliminates collisions. Correct Answer: Is used in wireless networks.	0 out of 1 points						
Question 32 Consider data delineation as implemented by various data link protocols. Answer this and the following four questions in such a context. Note that answers are NOT case sensitive. --- If the DATA 011111010 is to be transmitted via SDLC, how will this data (with no added surrounding 'markers') be transmitted? Selected Answer: 0111110010 Correct Answer: <table><tr><th>Evaluation Method</th><th>Correct Answer</th><th>Case Sensitivity</th></tr><tr><td> Exact Match </td><td>0111110010</td><td></td></tr></table>	Evaluation Method	Correct Answer	Case Sensitivity	Exact Match	0111110010		1 out of 1 points
Evaluation Method	Correct Answer	Case Sensitivity					
Exact Match	0111110010						
Question 33	0 out of 1 points						

The start of a token ring message is indicated by the octet JK0JK000.

Suppose the character ^ indicates a "high" signal and "_" (the underscore) indicates a "low" signal, the what will the actual signal look like that represents this octet. Assume the this octet starts with a "low" signal. (Include this "low" signal in your answer to account for all the signal levels used to represent this octet; do not insert any spaces in your answer.)

Selected Answer: ✖ _ ^ ^ ^ _ ^ ^ _ ^ _

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✔ Exact Match	_ ^ ^ ^ _ ^ ^ _ ^ _	

Question 340 out of 1 points

The payload of an Ethernet frame cannot exceed ____ octets.

Selected Answer: ✖ 155

Correct Answer: ✔ 1,500

Answer range +/- 0 (1500 - 1500)

Question 351 out of 1 points

The start of frame delimiter follows the preamble in an Ethernet frame. Use a sequence of zeroes and ones to indicate the value of this start of frame delimiter. Do not insert any spaces into your answer.

Selected Answer: ✔ 10101011

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✔ Exact Match	10101011	

Question 360 out of 1 points

Suppose we want to create a data link protocol that will only ever carry musical data. We sample the analogue signal 44100 per second and represent each sample as a 16-bit value. The music is in stereo, so two samples are taken at a time. We will use FFFF (hex) to delimit both the start and the end of the packet. No error checking or other headers are added. Whenever an FFFF occurs in the DATA packet, we change it to FFFE. This change should be hardly audible. Assume that signals are evenly spread so that combinations of the 16 bit values occur with equal frequency. (This is a rather pessimistic assumption.) Under this assumption, how many milliseconds of music will be changed in a frame that contains exactly 60 minutes of music?

Selected Answer: ✖ [None Given]

Correct Answer: ✔ 549,316

Answer range +/- 1 (549315 - 549317)

Question 371 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

000100011110

Selected Answer: ✔ 9

Correct Answer: ✔ 9

Answer range +/- 0 (9 - 9)

Question 381 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

010011001001

Selected Answer: ✔ 1

Correct Answer: ✔ 1

Answer range +/- 0 (1 - 1)

Question 391 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

000110110001

Selected Answer: ✔ 3

Correct Answer: ✔ 3

Answer range +/- 0 (3 - 3)

Question 401 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

001000001101

Selected Answer: ✔ 12

Correct Answer: ✔ 12

Answer range +/- 0 (12 - 12)

Question 411 out of 1 points

This question tests whether you understand Hamming code as it has been explained in class.

You use a Hamming code capable of correcting single bit errors. Bits are numbered from right to left. Even parity is used. You receive the character below.

Indicate in the appropriate space which bit has changed by only writing a number from 1 to 12. If no bit has changed, indicate it by writing 0.

101111101010

Selected Answer: 0
Correct Answer: 0
Answer range +/- 0 (0 - 0)

Question 42

1 out of 1 points

Assume 133.133.0.0 is a class B IPv4 address that is used by your company. You are given the task to subnet it such that at least 100 subnets are supported, with support for at least 500 computers on each subnet. Use this information for this question, as well as the four questions that follow.

What netmask will your company use internally after the subnetting?

Selected Answer: 255.255.254.0
Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
Exact Match	255.255.254.0	

Question 43

0 out of 1 points

Consider the address 133.133.133.133. What will its host number be?

Selected Answer: 34181
Correct Answer: 389
Answer range +/- 0 (389 - 389)

Question 44

0 out of 1 points

Consider subnet 40. The its broadcast address will be of the form 133.133.v.w

What is the value of v?

Selected Answer: 79
Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
Exact Match	81	

Question 45

0 out of 1 points

Consider the address 133.133.133.133. On which subnet will it be located?

Selected Answer: 67
Correct Answer: 66
Answer range +/- 0 (66 - 66)

Question 46

0 out of 1 points

Consider subnet 40. The its network address will be of the form 133.133.x.y/z.

What is the value of x?

Selected Answer: 78
Correct Answer: 80
Answer range +/- 0 (80 - 80)

Question 47

1 out of 1 points

In the video on IPv6 I stated that the localhost address is "all zeroes (::)". This was wrong. The localhost address is the number 1 (in decimal, binary, hexadecimal, octal and the vast majority of number systems you ought to care about). So, what is the correct localhost address, expressed in colon-hexadecimal notation with zero suppression?

(Hint: It is NOT :: as I said in the video.)

Selected Answer: ::1
Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
Exact Match	::1	

Question 48

1 out of 1 points

An IPv6 routing prefix consists of ____ bits.

Selected Answer: 64
Correct Answer: 64
Answer range +/- 0 (64 - 64)

Question 49

1 out of 1 points

The host number of an IPv6 address consists of ____ bits.

Selected Answer: 64
Correct Answer: 64
Answer range +/- 0 (64 - 64)

Question 50

1 out of 1 points

The subnets used by an organisation who uses an fd00::/8 "private" address

Selected Answer: May be located on different premises, because the subnet is included in the routing prefix.
Correct Answer: May be located on different premises, because the subnet is included in the routing prefix.

Question 51

0 out of 1 points

The command
ping ::ffff:1.2.3.4

will be handled by the ... at the origin site.

Selected Answer:  IPv6 stack

Correct Answer:  IPv4 stack

Question 52

0 out of 0 points

Have you cheated while writing this test?

Selected Answer:  False

Correct Answer:  False

Wednesday, July 1, 2020 5:13:57 PM SAST

← OK